

IOT-BASED SYSTEMS **(UEC-640)**

Project Title: VocalHand: AI-Driven Assistive Glove for Real-Time Sign Language Translation using TinyML and Predictive NLP

Team Members:

1. Pranav Sharma (102306061)
2. Mehak Goel (102306084)
3. Mahak Vijay (102306089)
4. Tanvi Gupta (102306092)
5. Sneha Arora (102306096)

Abstract

Many individuals with speech and hearing impairments face difficulties in daily communication because sign language is not commonly understood by the general public. This project aims to address this problem by developing a smart assistive glove that converts hand gestures into spoken words while also supporting two-way communication.

The glove uses sensors to detect finger bending and hand movements. A lightweight TinyML-based artificial intelligence model embedded in the glove recognises these gesture patterns and sends the corresponding output to a smartphone application. To make communication faster and more natural, the application includes next-word prediction, similar to smartphone keyboards, which helps users form complete sentences using fewer gestures.

Overall, the system works not only as a sign-language translator but also as an intelligent communication aid. It is low-cost, wearable, and easy to use making it suitable for people with **speech impairments** as well as **hearing impairments**.

Novelty of the Proposed Work

What Existing Papers Have Done:

Most existing systems use simple if-then rule-based logic, for example, “if a finger is bent, display a particular letter.” These systems generally use expensive commercial sensors and large circuit boards that need to be connected to a computer. Gesture recognition is limited to one letter at a time. Hand movement, tilting, and orientation are usually not detected.

Limitations or Gaps in Existing Work:

- **Slow Communication:** Spelling words letter by letter takes too much time for real-world conversations.
- **High Cost:** Professional-grade sensors are expensive and unaffordable for regular users.
- **Low Accuracy:** Simple rule-based logic becomes inaccurate if the glove slips or the user’s hand shakes.
- **One-Way Communication:** Existing systems help the speech-impaired person speak but do not help them understand responses from others.

Hardware-Level Novelty:

- **3D Motion Sensing:** An **MPU6050** sensor is added to detect hand movement, tilting, and orientation, similar to an inner ear for the glove.
- **Low-Cost DIY Sensors:** **Velostat**, a low-cost conductive material, is used instead of expensive commercial sensors, making the system affordable.
- **All-in-One Compact Design:** An **Arduino Nano** is used to keep the electronics small, lightweight, and suitable for wrist-mounted use.

Software and Algorithm-Level Novelty:

- **AI-Based Learning (TinyML):** A TinyML-based **neural network** is trained to recognise hand gesture patterns instead of using fixed rules, allowing accurate performance even with hand tremors.
- **Smart Prediction (Next-Word Prediction):** The mobile application predicts the next likely word. For example, after signing “I want” the system may suggest “water” or “food” reducing effort.
- **Two-Way Communication:** The application can convert spoken input from a hearing person into text, allowing the speech-impaired user to read and respond.

Impact of the Proposed Changes:

The proposed system enables faster and more natural communication. It performs well under real-world hand movements, unlike traditional systems. The use of **DIY hardware** and **AI** reduces the overall cost by nearly **90%** while maintaining reliable performance. This makes the system an affordable and life-improving assistive tool for individuals with speech and hearing impairments.